

D311

Microbiology with Lab: A Fundamental Approach

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student isolates bacteria from a kitchen sponge and performs a Gram stain. The colonies were creamy white and smelled slightly sweet. Under the microscope, the cells are purple cocci in clusters. What does the purple color show about the cell wall?

- A. A thin peptidoglycan layer with an outer membrane.
- B. A thick peptidoglycan layer that kept the crystal violet.
- C. No cell wall at all.
- D. A sugar-rich capsule, based on the sweet smell.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

After a Gram stain, a rod-shaped bacterium appears pink. Which structure would you expect it to have?

- A. No peptidoglycan whatsoever.
- B. An outer membrane containing lipopolysaccharide.
- C. A nucleus.
- D. A thick peptidoglycan wall.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A lab sample from a patient with a skin rash contains budding oval cells that grew best at room temperature on a slightly acidic agar. The patient recently returned from a beach vacation. How is this organism classified?

- A. Eukaryote, because yeasts have a nucleus and membrane-bound organelles.
- B. Virus, because it caused a rash.
- C. Prokaryote, because it is a single cell.
- D. A marine bacterium picked up at the beach.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A broth culture starts with 500 bacterial cells. The species doubles every 20 minutes at 37 °C, and the flask was sealed with a foam plug. Assuming ideal growth, how many cells are present after 2 hours?

- A. 2,000
- B. 64
- C. 3,000
- D. 32,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A culture grows from 1,000 cells to 16,000 cells in 2 hours during exponential growth. What is the generation time?

- A. 30 minutes
- B. 4 minutes
- C. 7.5 minutes
- D. 60 minutes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A daycare has an outbreak of a stomach virus spread by contaminated hands and surfaces. The center serves 40 children and recently repainted its playroom. Which action best breaks the chain of infection?

- A. Placing all children in N95 respirators.
- B. Giving all children antibiotics.
- C. Strict handwashing with soap and water plus routine surface disinfection.
- D. Closing the repainted playroom.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A nursing student receives the hepatitis B vaccine series before starting clinicals. She works part-time at a pharmacy and exercises daily. Weeks later, blood tests show she has made anti-HBs antibodies. What type of immunity is this?

- A. Artificially acquired passive immunity
- B. Artificially acquired active immunity
- C. Naturally acquired active immunity
- D. Innate immunity

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A newborn is protected against measles for several months without ever being vaccinated or infected. Where did this protection come from?

- A. An antibody injection: artificially acquired passive immunity.
- B. Lifelong immunity present at birth.
- C. Maternal antibodies that crossed the placenta: naturally acquired passive immunity.
- D. The baby's own memory B cells: active immunity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A lab tests an unknown isolate by disk diffusion. The tetracycline zone is 22 mm (susceptible at 19 mm or more), and the erythromycin zone is 9 mm (resistant at 13 mm or less). The patient says an older relative once did well on erythromycin. Which drug do the results support?

- A. Tetracycline
- B. Erythromycin, because a smaller zone means a stronger drug
- C. Erythromycin, because it worked for a relative
- D. Either one, since both made a zone

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Drug X has an MIC of 0.5 µg/mL against an isolate, and drug Y has an MIC of 8 µg/mL. Which statement is correct?

- A.** MIC measures the concentration that kills all cells.
- B.** Drug Y is more potent because its number is higher.
- C.** They are equally effective.
- D.** Drug X inhibits growth at a lower concentration than drug Y.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A college student has a runny nose, sore throat, and mild cough for three days, and testing confirms a rhinovirus. He has a big exam on Friday and asks for a Z-pack like his roommate got last month. What is the best explanation?

- A.** Antibiotics will shorten the illness if started early.
- B.** The roommate's prescription shows antibiotics work for colds.
- C.** Antibiotics target bacterial structures like cell walls and 70S ribosomes, which viruses lack, so they will not help.
- D.** Antibiotics never have any use in a patient with a virus.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A clinic must reprocess reusable surgical instruments that may be contaminated with Clostridium spores. The clinic recently switched to a new cleaning supply vendor. Which method is required?

- A.** Any product from the new vendor labeled as a cleaner.
- B.** Scrubbing with antiseptic skin soap.
- C.** Wiping with 70% alcohol.
- D.** Sterilization, such as autoclaving at 121 °C under pressure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A student isolates bacteria from a kitchen sponge and performs a Gram stain. ~~The colonies were creamy white and smelled slightly sweet.~~ Under the microscope, the cells are purple cocci in clusters. What does the purple color show about the cell wall?

Target word: wall structure from stain color **Decides:** Purple = crystal violet retained = gram-positive.

A	Inverted operation	That describes gram-negative cells, which stain pink.
B	Correct	Gram-positive walls retain the primary stain.
C	Adjacent category	Wall-less bacteria (e.g., mycoplasma) do not Gram stain well.
D	Camouflage	Odor and colony color do not determine Gram reaction.

2. Answer B Concept 1 · transfer

After a Gram stain, a rod-shaped bacterium appears pink. Which structure would you expect it to have?

Target word: reverse: color to structure **Decides:** Pink = gram-negative.

A	Absolute language	Gram-negatives have a thin peptidoglycan layer.
B	Correct	Gram-negatives have an LPS outer membrane over a thin wall.
C	Confusion set	No bacterium has a nucleus.
D	Inverted operation	That is gram-positive.

3. Answer A Concept 2

A lab sample from a patient with a skin rash contains budding oval cells that grew best at room temperature on a slightly acidic agar. ~~The patient recently returned from a beach vacation.~~ How is this organism classified?

Target word: cell type **Decides:** Budding oval cells = yeast, a fungus.

A	Correct	Fungi are eukaryotes.
B	Confusion set	Viruses do not bud as free cells on agar.
C	Adjacent category	Being single-celled does not make an organism prokaryotic.
D	Camouflage	Travel history does not decide cell type; budding does.

4. Answer D Concept 3

A broth culture starts with 500 bacterial cells. The species doubles every 20 minutes at 37 °C, ~~and the flask was sealed with a foam plug.~~ Assuming ideal growth, how many cells are present after 2 hours?

Target word: population after time **Decides:** 2 hours / 20 minutes = 6 generations.

A	Wrong stem number	Uses 2 hours as 2 generations (500×2^2).
B	Dropped element	2^6 without the starting 500 cells.
C	Inverted operation	500×6 treats growth as linear instead of doubling.
D	Correct	$500 \times 2^6 = 500 \times 64 = 32,000$.

5. Answer A Concept 3 • transfer

A culture grows from 1,000 cells to 16,000 cells in 2 hours during exponential growth. What is the generation time?

Target word: solve for generation time **Decides:** $16,000 / 1,000 = 16 = 2^4$, so 4 generations.

A	Correct	120 minutes / 4 generations = 30 minutes.
B	Adjacent category	4 is the number of generations, not the time per generation.
C	Wrong stem number	120 / 16 divides by the fold increase, not the number of doublings.
D	Dropped element	Counts only two doublings.

6. Answer C Concept 4

A daycare has an outbreak of a stomach virus spread by contaminated hands and surfaces. ~~The center serves 40 children and recently repainted its playroom.~~ Which action best breaks the chain of infection?

Target word: break the chain **Decides:** Contact route: hands and surfaces.

A	Confusion set	Respirators target airborne spread, not contact.
B	Adjacent category	Antibiotics do not treat or prevent viral illness.
C	Correct	Breaks the contact mode of transmission.
D	Camouflage	Fresh paint has nothing to do with transmission.

7. Answer B Concept 5

A nursing student receives the hepatitis B vaccine series before starting clinicals. ~~She works part time at a pharmacy and exercises daily.~~ Weeks later, blood tests show she has made anti-HBs antibodies. What type of immunity is this?

Target word: type of immunity **Decides:** Her own body made the antibodies after a vaccine.

A	Inverted operation	Passive means receiving ready-made antibodies.
B	Correct	A vaccine prompts the body to make its own antibodies.
C	Adjacent category	Natural active immunity comes from actual infection.
D	Confusion set	Innate defenses are nonspecific and do not make targeted antibodies.

8. Answer C Concept 5 • transfer

A newborn is protected against measles for several months without ever being vaccinated or infected. Where did this protection come from?

Target word: reverse: identify passive immunity **Decides:** No vaccine or infection; antibodies came from the mother.

A	Adjacent category	No injection was given.
B	Absolute language	Maternal antibodies fade over months.
C	Correct	Ready-made antibodies from the mother; temporary.
D	Inverted operation	The baby had no exposure to form memory cells.

9. Answer A Concept 6

A lab tests an unknown isolate by disk diffusion. The tetracycline zone is 22 mm (susceptible at 19 mm or more), and the erythromycin zone is 9 mm (resistant at 13 mm or less). ~~The patient says an older relative once did well on erythromycin.~~ Which drug do the results support?

Target word: which drug **Decides:** Tetracycline's zone meets the susceptible breakpoint.

A	Correct	22 mm meets the 19 mm susceptible breakpoint.
B	Inverted operation	A small zone means the bacteria grew close to the disk: resistant.
C	Camouflage	Another person's history does not predict this isolate's susceptibility.
D	Absolute language	Zone size must be compared with each drug's breakpoints.

10. Answer D Concept 6 · transfer

Drug X has an MIC of 0.5 µg/mL against an isolate, and drug Y has an MIC of 8 µg/mL. Which statement is correct?

Target word: reading MIC **Decides:** Lower MIC = less drug needed to inhibit growth.

A	Confusion set	That is the MBC; MIC measures inhibition.
B	Inverted operation	A higher MIC means more drug is required.
C	Dropped element	Ignores the 16-fold difference in MIC.
D	Correct	0.5 is less than 8.

11. Answer C Concept 7

A college student has a runny nose, sore throat, and mild cough for three days, and testing confirms a rhinovirus. ~~He has a big exam on Friday and asks for a Z-pack like his roommate got last month.~~ What is the best explanation?

Target word: antibiotics for a virus **Decides:** The cause is a virus.

A	Inverted operation	They have no effect on viral replication.
B	Camouflage	Someone else's prescription does not decide treatment.
C	Correct	Viruses have no bacterial targets.
D	Absolute language	A secondary bacterial infection may need them.

12. Answer D Concept 8

A clinic must reprocess reusable surgical instruments that may be contaminated with Clostridium spores. ~~The clinic recently switched to a new cleaning supply vendor.~~ Which method is required?

Target word: method that kills spores **Decides:** Endospores require sterilization.

A	Camouflage	The vendor is irrelevant; the method must be sterilization.
B	Confusion set	Antiseptics are for living tissue.
C	Adjacent category	Alcohol is a disinfectant/antiseptic and does not kill spores.
D	Correct	Sterilization kills all microbes including endospores.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Gram stain and cell wall	Gram-positive cells have a thick peptidoglycan wall that keeps crystal violet (purple); gram-negative cells have a thin wall plus an outer membrane and stain pink with safranin.	Reversing the colors or thinking gram-negatives have the thicker wall.	A color after staining, or a question about wall structure.
2	Prokaryote vs. eukaryote	Prokaryotes lack a nucleus and membrane-bound organelles and have 70S ribosomes; eukaryotes (fungi, protozoa, humans) have a nucleus and 80S ribosomes.	Classifying yeast or protozoa as prokaryotes.	Presence or absence of a nucleus.
3	Exponential bacterial growth	Population = starting number $\times 2^{(\text{number of generations})}$; generations = time / generation time.	Adding cells linearly instead of doubling.	A starting count, a doubling time, and an elapsed time.
4	Chain of infection and transmission	Infection needs agent, reservoir, portal of exit, mode of transmission, portal of entry, and susceptible host; breaking any link stops spread, and hand hygiene breaks contact transmission.	Matching the wrong precaution to the mode (e.g., airborne vs. droplet).	How the organism travels from one person to another.
5	Active vs. passive immunity	Active immunity comes from the body making its own antibodies (infection or vaccine) and lasts; passive immunity comes from receiving ready-made antibodies (maternal, antibody injection) and is short-lived.	Calling a vaccine passive or antibody injections active.	Who made the antibodies.
6	Antimicrobial susceptibility testing	In disk diffusion, a zone at or above the drug's susceptible breakpoint means susceptible; a lower MIC means less drug is needed to inhibit growth.	Thinking a small zone or high MIC means a better drug.	Zone sizes in mm or MIC values in $\mu\text{g/mL}$.
7	Viruses are not cells	Viruses are obligate intracellular agents without ribosomes or cell walls, so antibiotics do not treat them.	Prescribing antibiotics for a viral infection.	A viral cause plus a question about treatment.
8	Sterilization vs. disinfection vs. antisepsis	Sterilization kills all microbes including endospores (e.g., autoclave at 121 °C); disinfection reduces microbes on objects; antisepsis reduces microbes on living tissue.	Assuming disinfectants or alcohol kill endospores.	Surgical instruments, surfaces, or skin, and whether spores matter.

Course scope (public description): Microbiology with Lab: A Fundamental Approach explores the science that microorganisms are everywhere, and they have positive and negative effects on the community. The course

examines the structure and function of microorganisms, disease transmission and progression, and immune responses and other interventions, and it identifies key global diseases. The course consists of an introduction and four major sections. Each section includes learning opportunities through readings, videos, and other relevant resources. Assessment activities with feedback also provide opportunities for students to check their learning, practice, and show how well they understand course content. To assist students in developing an applied, evidence-based understanding of microbiology, this course integrates several lab experiments to help determine the specific characteristic of an unknown microbial sample and a treatment plan. Because the course is self-paced, students may move through the material as quickly or as slowly as needed to gain proficiency in the four competencies that will be covered in the final assessment. Students who have no prior knowledge of or experience with this topic can expect to spend 48–60 hours on the course content. There are no prerequisites for this course.